

Knowledge Graphs and Distributional Readability of Arabic Documents

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Abstract

This paper addresses automatic readability assessment of Arabic documents. It compares classical readability formulae, fine-tuned Arabic encoders, zero-shot large language models and a Co-occurrence Graphs for Arabic Readability (CoGAR). CoGAR, interpretable and LLM-free, represents a document as a *co-occurrence network* over content lemmas and use *topology* features such as density, clustering, path length and degree entropy to predict readability. Evaluated on the BAREC corpus over five ordinal levels, the method reaches a quadratic weighted kappa (QWK) of 0.756, on par with a sparse 35B-parameter LLM and well above the surface formulas Flesch and OSMAN. An ablation shows that the signal lies in the graph’s *shape* rather than size, and that alternative graphs based on relations between named-entity carry almost no readability difficulty signal. We further distill a frontier LLM (Gemini) into a small Arabic encoder (NeoAraBERT) by silver labelling 30k chunks, reaching 0.714 QWK. We show that *stacking* an LLM with CoGAR improves on the LLM alone, reaching QWK 0.839 with GPT-4.1 and 0.852 with the open-weight Qwen-35B, our best result. The two signals are complementary and the gain does not depend on a proprietary model. Finally, we analyze how eight methods including CoGAR distribute reading levels across seven Arabic text sources ranging from children books to expert research manuscripts showing how the methods align with intuitive difficulty.

1 Introduction

Curation of text readability difficulty levels is central to producing learning materials, grading curricula, simplifying documents, and training large language models (LLMs) (Zhang et al., 2026). For Arabic, this task is made harder by rich morphology and optional diacritics, and it has long relied on formulae that range over character counts,

and difficult word frequencies (Dawood, 1977; Al-Tamimi et al., 2014; El-Haj and Rayson, 2016). Recent neural network and LLM methods open new possibilities, but they are opaque and costly. Even when successful, it remains unclear how much of the difficulty signal is *structural* versus related to other features such as length and aggregate word difficulty signals.

This paper refers to any graph built based on content extracted from a document as a knowledge graph (KG). We study two such graphs: word *co-occurrence* graphs, whose nodes are content lemmas and whose edges link words that appear close together, together with semantic *entity-relation* graphs, the conventional knowledge graph of named entities and typed relations. Our central finding is that the co-occurrence graph carries almost all of the readability signal, while the entity-relation graphs we extracted were not informative. CoGAR alone, which is interpretable and LLM-free, performs at the level of a 35B-parameter LLM. Stacking CoGAR with an LLM achieves the best score including comparison with larger LLMs.

The paper makes the following contributions.

- A co-occurrence **network-topology** method for Arabic readability that is interpretable and LLM-free, yet competitive with large LLMs (QWK 0.756 ± 0.001 over repeated cross-validation).
- A controlled **ablation** isolating which graph signals carry difficulty: topology dominates volume, while named-entity and relation graphs provide no additional value.
- A **systematic comparison** on BAREC (Elmadani et al., 2025b) of readability formulas, two fine-tuned Arabic encoders, three LLMs and the graph, under a single ordinal evaluation.

- A **knowledge-distillation** pipeline turning a frontier LLM (Gemini (Gemini Team, Google DeepMind, 2026)) into a small encoder (NeoAraBERT (Abou Chakra et al., 2026)), reaching QWK 0.714.
- A **stacking** of an LLM and CoGAR that reaches QWK 0.839 with GPT-4.1 and 0.852 with the open-weight Qwen-35B, showing the two signals are complementary and that the gain does not depend on a proprietary model.
- A **cross-source distributional study** of eight methods over seven Arabic categorical text types (children books, mystery fiction, . . . , Academic research manuscripts) that map to conventional age/educational groups, on unlabeled data.

Our code is publicly available.¹

2 Related Work

Arabic-specific readability factors. Several features directly affect the readability of Arabic text. Two of them are particularly useful for estimating the difficulty level of a sentence: diacritics and morphology. Diacritics help disambiguate intended semantics, role in a sentence, and pronunciation. When omitted young readers may struggle to read Arabic, which is why books intended for children are usually heavily vocalized. Arabic morphology relies on a root-and-pattern system: the root is a set of three, four or exceptionally five letters carrying a central meaning or concept. The root system helps: knowing a root often reveals the meaning of a new word. Prefixes, suffixes and infixes apply to the same word, so a beginner struggles to separate morphemes.

Syntactic complexity and lexical difficulty also matter, though neither appears in surface counts. These properties make estimating the difficulty of an Arabic text non-trivial and explain why dedicated readability measures have been developed.

The earliest readability formulas. The study of readability is not recent. The earliest approaches rely on mathematical formulas based on word and sentence length. Flesch (1948) proposed the Flesch Reading Ease formula, one of the oldest:

$$\text{FRE} = 206.835 - 1.015 \text{ ASL} - 84.6 \text{ ASW},$$

¹<https://github.com/U4RASD/arabic-readability-cogar>

where ASL and ASW are the average sentence length (words per sentence ratio) and average number of syllables per word. The score ranges from 0 to 100: the higher it is, the easier the text is to read. The U.S. Navy adopted this into the Flesch–Kincaid formula (Kincaid et al., 1975) to indicate the school grade level required to understand a text:

$$\text{FK} = 0.39 \text{ ASL} + 11.8 \text{ ASW} - 15.59.$$

Other formulas followed, such as the Gunning Fog Index (Gunning, 1969) and SMOG (McLaughlin, 1969). Several studies (Bruce et al., 1981; Redish, 2000) critically examined the limitations and degrees of accuracy of the existing readability formulas. These formulas do not transfer directly from English to Arabic, which motivated the development of language-specific formulas.

Arabic readability formulas. Among the first formulas dedicated to Arabic, Dawood (1977) proposed a formula based on three textual features: AWL (average word length), ASL, and WF (word frequency):

$$\begin{aligned} \text{Score} = & -0.0533 \text{ AWL} - 0.2066 \text{ ASL} \\ & + 5.5543 \text{ WF} - 1.0801. \end{aligned}$$

Later, Al-Tamimi et al. (2014) published the Arabic Automatic Readability Index (AARI), combining the number of characters (NoC), ASL, and AWL:

$$\text{AARI}_{\text{base}} = 3.28 \text{ NoC} + 1.43 \text{ AWL} + 1.24 \text{ ASL},$$

. Regression then maps the results to school grade levels:

$$\text{Grade} = (\text{AARI}_{\text{base}} + 472.42)/1046.3.$$

El-Haj and Rayson (2016) proposed the OSMAN formula, which evaluates readability with or without diacritics and introduces a “Faseeh” factor (the letters ء, ئ, ؤ, ذ, ظ and the final sequences و and ن):

$$\begin{aligned} \text{OSMAN} = & 200.791 - 1.015 \frac{A}{B} \\ & - 24.181 \frac{C+D+G+H}{A}, \end{aligned}$$

where A is the total number of words, B the number of sentences, C the number of hard words (> 5 letters), D the number of syllables per word, G the number of complex words (> 4 syllables), and H the number of Faseeh words. All these formulas rely on surface cues (lengths, counts); to capture finer aspects of difficulty. Subsequent work turned to richer linguistic features and machine learning.

Feature-based and machine-learning approaches. We distinguish here between Arabic as a first language (L1, native speakers) and Arabic as a second language (L2, foreign learners); the difficulty of a text is not measured the same way for these two audiences. [Saddiki et al. \(2015\)](#) sought to determine the performance achievable with a minimal set of easy-to-extract features: using a GLOSS corpus of 251 texts spread over five ILR (Interagency Language Roundtable proficiency) levels and 35 low-complexity features, they reported a maximum accuracy of 59.76% for five classes, rising to 73.31% when reducing the problem to three classes. [Saddiki et al. \(2018\)](#) examined a much larger set of features on a GLOSS-based corpus of 545 Modern Standard Arabic texts; with four classes, their best L2 accuracy reached 72.4%. [Nassiri et al. \(2020\)](#) released the MoSAR corpus (Modern Standard Arabic Readability), a resource for training, optimizing, and evaluating readability approaches for L1 learners; they later used it ([Nassiri et al., 2022](#)) to assess the benefit of combining classical features with more sophisticated ones, reaching an accuracy of about 78%. [Khallaf and Sharoff \(2021\)](#) classified the difficulty of sentences for L2 learners by combining sentence embeddings with classical linguistic features, reporting F-scores of 0.80 and 0.75 with Arabic-BERT and XLM-R respectively.

Arabic readability resources. Several resources now support fine-grained Arabic readability. The SAMER project released a leveled readability lexicon of 26,000 lemmas for Modern Standard Arabic ([Al Khalil et al., 2020](#)), which later enabled the SAMER Corpus ([Alhafni et al., 2024](#)), the first manually annotated Arabic parallel corpus for text simplification, covering 159K words with readability annotations at both the document and word levels. [El-Haj et al. \(2024\)](#) introduced DARES, a dataset of 13,335 instances from Saudi school textbooks, annotated both at a coarse educational level and at the level of individual grades. The BAREC corpus ([Elmadani et al., 2025b](#); [Habash et al., 2025](#)) that we use here annotates sentences on a 19-level scale, and has since been extended to BAREC-10M ([Elmadani et al., 2026](#)), which reaches 10 million words across 45 sub-corpora annotated for morphology, syntax and readability.

Arabic readability modeling. [Liberato et al. \(2024\)](#) compared approaches ranging from rule-based methods to Arabic pretrained language models, reaching a macro F1 of 87.9 at the fragment level. The BAREC corpus paper ([Elmadani et al., 2025b](#)) benchmarks several models on its 19-level scale and releases the resulting classifiers, two of which we use as baselines. The BAREC Shared Task 2025 ([Elmadani et al., 2025a](#)) extended this benchmarking to sentence-level and document-level tracks, drawing 22 registered teams, with the winning system reaching 87.5 QWK at the sentence level.

Structural and neural approaches. Beyond surface features, some work exploits the structure of the text or neural representations, often for languages other than Arabic. [Feng et al. \(2010\)](#) show that, among discourse features, only entity-density features have a significant impact on predicting the readability of texts for elementary-school students (for English); this finding motivates examining text structure, which we approach in this work through word co-occurrence networks. [Qiu et al. \(2021\)](#) propose incorporating linguistic features into neural network models by learning dense syntactic embeddings, and conclude that this approach can complement BERT to substantially improve automatic readability assessment.

Large language models. Since the emergence of LLMs such as GPT ([OpenAI, 2025](#)), Qwen ([Qwen Team, 2026](#)), and Gemini ([Gemini Team, Google DeepMind, 2026](#)), these models have demonstrated strong performance on many natural language processing tasks and across many languages, including Arabic ([Al-Khalifa et al., 2025](#)). Trained on massive corpora, they possess broad lexical, syntactic, and semantic knowledge, making them in principle capable of judging the difficulty of a text without task-specific training (zero-shot), from a simple instruction describing the level scale. Combined with the emergence of Arabic-specialized LLMs, this capability makes them natural candidates for readability assessment, an approach we evaluate in this work alongside classical methods.

Challenges and positioning. Despite these advances, Arabic readability assessment remains limited by the scarcity of fine-grained annotated corpora ([Nassiri et al., 2023](#)). Existing resources often rely on coarse grade-level labels (three or

four classes), and many Arabic text collections lack difficulty annotations altogether. In this work, we address this gap by evaluating on the fine-grained BAREC corpus (Elmadani et al., 2025b) and by analysing, without gold labels, seven Arabic text sources spanning children’s stories, textbooks, news, and research books.

3 Data

Ground truth (BAREC). We use the BAREC corpus (Elmadani et al., 2025b; Habash et al., 2025) as ground truth for supervised evaluation. We build a five-level evaluation set of 1,000 paragraphs, 200 per level (levels 0 to 4, easiest to hardest). Our levels 1 to 4 correspond to BAREC levels 10, 12, 14 and 16, a regular subsampling of the BAREC scale rather than its first four levels. Each paragraph accumulates consecutive sentences of a single BAREC document, in document order, until a window of 20 to 60 words is reached (mean ≈ 42 words, 97% within the window). Its level is that of its hardest sentence, following the BAREC aggregation rule (Elmadani et al., 2025b). This leaves the low levels almost empty, since a paragraph stays low only if all its sentences are: levels 1 to 6 yield between 0 and 63 paragraphs, far below the 200 needed per class. Levels 10, 12, 14 and 16 provide abundant material with a regular spacing that keeps classes well separated.

Level 0, the easiest, does not come from BAREC: it consists entirely of short sentences from a first-grade Arabic-language textbook of the Saudi national curriculum (Ministry of Education, Kingdom of Saudi Arabia, 2024), with a mean length of 4.4 words per sentence. We use this source rather than the lowest BAREC level because, upon inspection, the easiest BAREC sentences are dominated by non-sentential fragments (titles, author names, page numbers and other editorial metadata) that do not constitute coherent reading material. These sentences are not annotated under the BAREC guidelines and we claim no formal equivalence with its scale. As an indirect check, the strongest released CAMEL classifier places 97.5% of them below BAREC level 10, which corresponds to our level 1, with a median of level 6.

Multi-source corpus. For a practical distributional study (Section 7) we collect 4,200 chunks, 600 from each of seven categorically different sources: children’s stories, detective stories,

middle-school textbooks, high-school textbooks, Arabic Wikipedia, newspaper articles, and academic research manuscripts. While these documents are not labeled on difficulty, each of them reflects a categorically suitable age group and educational level. Intuition is the distribution across readability levels should reflect the categorical difference between these corpora.

We segment each of these sources at sentence boundaries into the same 20–60 word window. We use these chunks to compare how methods *distribute* levels.

Silver corpus for distillation. For knowledge distillation (Section 4) we label 30,000 chunks with a frontier LLM (Gemini-3.5-flash), which serve as silver training data for a small Arabic encoder. Appendix C documents the provenance and versions of every dataset and model used in this work.

4 Methods

Readability is an *ordinal* task: predicting level 3 for a level 4 text is a smaller error than predicting level 0. We therefore score every method with the quadratic weighted kappa (QWK) (Cohen, 1968), which penalises a large deviation more than a small one.

Readability formulas. We apply three formulas: OSMAN (El-Haj and Rayson, 2016), AARI (Al-Tamimi et al., 2014) and a version of Flesch reading ease that adapts for English syllabification. Each continuous score is mapped to the five levels by rank binning into equal-frequency groups, oriented so that higher levels correspond to harder texts. Flesch and OSMAN are reading-*ease* scores, where a higher raw value means an easier text, so their bins are reversed. AARI is a grade-level index that already increases with difficulty.

Fine-tuned Arabic encoders. As supervised neural baselines we use two publicly available readability classifiers of CAMEL Lab (Elmadani et al., 2025b), applied directly to our paragraphs: `readability-arabertv02-word-CE`, which takes raw text, together with `readability-arabertv2-d3tok-reg`, the strongest released variant, which takes the D3Tok morphological tokenization as input and uses a regression head. Both models were already fine-tuned on BAREC, the same corpus our

levels 1–4 are drawn from, which gives them a strong in-domain advantage as baselines.

LLMs as zero-shot raters. We prompt three LLMs: GPT-4.1 (OpenAI, 2025), Qwen-35B (Qwen3.5-35B-A3B) (Qwen Team, 2026), and Fanar (Fanar-C-2-27B) (TEAM et al., 2026), with a fixed rubric that defines the five reading levels and asks for a single integer (Appendix D). The same rubric is used to produce the Gemini silver labels.

CoGAR: Co-occurrence Graphs for Arabic Readability. *Intuition.* We hypothesize that the difficulty of a text is reflected in the *structure* of how its words connect, not only in surface statistics such as length or frequency: a harder text weaves a denser, more interconnected network of concepts, whereas an easier text yields a simpler, more linear one. We capture this structure with a word co-occurrence network.

Graph construction. Each document is analysed with SinaTools 1.0.9 (Hammouda et al., 2024) to obtain, for every word, its lemma, root, part of speech, and frequency. We keep only *content lemmas* (nouns, verbs, adjectives) and discard function words. We then build an *undirected* graph whose nodes are these lemmas and whose edges link two lemmas that co-occur within a window of two consecutive content words.

Features. From this graph we extract two types of metrics: *volume* (number of nodes and edges), which merely reflect text size, and *topology*, which describes the shape of the network independently of its size: density, average clustering coefficient (how interconnected a word’s neighbours are), average shortest-path length and diameter (how far apart concepts are), average degree, degree entropy (how uneven word connectivity is), and assortativity (whether highly connected words link to one another). In total we compute 40 features in four families: lexical (8: lemma and root diversity, length, rarity via frequency), co-occurrence (12: 2 volume and 10 topology), named entities (5: diversity and density over the 21 Wojoood (Jarrar et al., 2022) categories) and an entity-relation graph (15: relation counts, types and topology). Appendix A lists all of them individually.

Entity–relation graph. We also build a directed entity–relation graph for each document, a standard knowledge graph of entities and the relations between them. Using SinaTools, we extract the named entities (the same 21 entity categories as above) together with the semantic relations that

hold between them; each entity becomes a typed node and each extracted relation becomes a directed, labelled edge between two entities. From this graph we derive the 15 relation features: the number of relations, the number of distinct relation types and their entropy, and the same topology measures as for the co-occurrence graph (density, clustering, path length, degree entropy, assortativity). This graph encodes *which entities are related to which*, in contrast to the co-occurrence graph, which only encodes which content words appear close together.

Model and training. As readability is ordinal, CoGAR predicts the level with an ordinal regressor (LogisticAT (Pedregosa et al., 2017)), which respects the ordering of the levels. All supervised models are trained and evaluated by stratified five-fold cross-validation on our 1,000-paragraph set (BAREC levels 1–4 together with the level-0 Saudi textbook): every paragraph receives its prediction from a fold in which it was not used for training (out-of-fold prediction). Cross-validation lets us evaluate on the whole set, which matters at this size, since rebuilding our paragraphs under the official BAREC splits would leave classes too imbalanced for the task. The CAMEL classifiers are applied to the same paragraphs. We verify stability by repeating the cross-validation over ten random seeds (reported in the robustness analysis below).

Ablation design. Finally, we evaluate each feature family *in isolation*, using the same ordinal model and the same cross-validation, to determine whether the difficulty signal comes from the graph’s *shape* (topology) or merely from its *size* (volume).

Knowledge distillation. We distill Gemini-3.5-flash (Gemini Team, Google DeepMind, 2026) into NeoAraBERT (Abou Chakra et al., 2026), a small Arabic encoder. The teacher labels 30,000 chunks on the five-level scale; the student is then fine-tuned on these silver labels for three epochs. We train two variants, one with a class-weighted cross-entropy loss and one with a focal loss. We report both. The focal variant transfers marginally better (0.714 against 0.708), a gap we do not treat as meaningful since the comparison was made on BAREC rather than on a held-out set. Notably, the student never sees any human annotation: it learns the readability scale entirely from the LLM’s silver labels, and is evaluated on the human-labelled BAREC set.

5 Experimental Setup

Our primary metric is the quadratic weighted kappa (QWK), introduced above. We also report exact accuracy, within-1 and within-2 accuracy, mean absolute error (MAE), and Spearman correlation. Supervised models use stratified five-fold out-of-fold cross-validation, for the graph we additionally repeat the cross-validation over ten random seeds to report mean and standard deviation.

6 Results

6.1 Main comparison

Table 1 reports all methods on BAREC, sorted by QWK. The strongest released CAMEL variant, *arabertv2-d3tok-reg*, obtains the highest score (0.860). As an encoder fine-tuned on the annotated BAREC sentences from which our levels 1 to 4 are drawn, it benefits from a strong in-domain advantage. Among the remaining systems, GPT-4.1 leads (QWK 0.784), although its margin over CoGAR is small and at the edge of significance (paired bootstrap, $+0.028$, $p = 0.05$). CoGAR follows closely: the topology variant reaches 0.756 ± 0.001 and the full feature set 0.750 ± 0.005 , statistically indistinguishable from the sparse Qwen3.5 model (35B total, 3B active, QWK 0.751, $p = 0.83$) and clearly above Fanar (0.702) and the *arabertv02-word-CE* classifier (0.694). The distilled NeoAraBERT (0.714) is competitive with Fanar and with the *arabertv02-word-CE* classifier. Flesch and OSMAN are reading-*ease* scores while AARI is a grade-level index. We bin all of them in the direction of increasing difficulty, so every reported value is positive: OSMAN 0.215 and Flesch 0.186, both far below every learned method, while AARI reaches 0.607. For every strong method, within-2 accuracy is close to 0.99, i.e. predictions are almost never off by more than two levels.

6.2 What in the graph carries the signal?

Table 2 compares feature families in isolation, all using the same ordinal model and cross-validation. The entity-relation graph is essentially uninformative (0.023) and named entities are weak (0.338): entity density follows the *topic*, not the difficulty:

²For the rank-binned formula methods, the predicted levels have a uniform marginal identical to the balanced gold distribution; under this condition quadratic weighted kappa reduces to the Spearman rank correlation, so the two columns coincide.

the hardest, most abstract texts contain almost no linked entities, whereas a co-occurrence graph exists for every text. The co-occurrence *volume* (size alone) reaches 0.682, but the *topology* rises to 0.756, showing that the difficulty signal is in the network’s *shape*, not its size. Crucially, adding AARI (explicit length) to the full feature set brings no meaningful gain (*aari+all* 0.752 vs. all features 0.750, well within one standard deviation): length is already encoded in the graph.

6.3 Feature importance

A random-forest importance over the full feature set confirms this picture: nine of the fifteen most important features are co-occurrence topology, and the top four are all graph-structural (*cooc_density* 0.070, *cooc_n_edges* 0.069, *cooc_avg_clustering* 0.063, *cooc_avg_path* 0.062), ahead of the strongest lexical feature (words per sentence, 0.061).

Sensitivity to the co-occurrence window. Table 6 in Appendix B reports the topology configuration for window sizes from one to five content words, over five seeds per value. The total range is 0.024, so the method is robust to this parameter. A window of one obtains a slightly higher score than our choice of two, a difference of 0.008 within that range, which indicates that we did not tune this parameter on the evaluation data.

6.4 Robustness

Repeating the cross-validation over ten seeds, co-occurrence topology reaches $\text{QWK} = 0.756 \pm 0.001$ (10 seeds $\times$ 5 folds) versus 0.750 ± 0.005 for all features. A paired bootstrap over the same seeds finds no significant difference between the two configurations ($p = 0.52$), so we do not claim that topology performs better. We adopt it as our method of choice because it is five times more stable and relies on four times fewer features while reaching the same level of performance.

6.5 Significance testing

To support the system-versus-system claims, we run a paired bootstrap over the ten cross-validation seeds. CoGAR topology is statistically indistinguishable from Qwen-35B (difference below 0.01, $p = 0.83$) and is numerically ahead of it. GPT-4.1 scores higher than CoGAR, but by a small margin at the edge of significance ($+0.028$, $p = 0.05$), which is why we avoid describing its lead as es-

Method	Exact	± 1	± 2	MAE	Spearman	QWK
CAMeL arabertv2-d3tok-reg	0.571	0.989	0.999	0.441	0.887	0.860
GPT-4.1 (zero-shot)	0.506	0.921	0.998	0.575	0.836	0.784
CoGAR (topology)	0.497	0.889	0.993	0.621	0.734	0.756 ± 0.001
Qwen-35B (zero-shot)	0.466	0.911	0.998	0.625	0.834	0.751
CoGAR (all features)	0.491	0.886	0.989	0.634	0.731	0.750 ± 0.005
NeoAraBERT (distilled, focal)	0.443	0.893	0.998	0.666	0.808	0.714
Fanar (zero-shot)	0.408	0.913	0.988	0.691	0.778	0.702
CAMeL arabertv02-word-CE	0.515	0.893	0.955	0.654	0.740	0.694
AARI	0.458	0.785	0.923	0.834	0.607	0.607
OSMAN	0.224	0.564	0.814	1.416	0.215	0.215
Flesch	0.245	0.583	0.813	1.404	0.186	0.186

Table 1: Main comparison on BAREC ($n = 1000$, five ordinal levels), sorted by quadratic weighted kappa (QWK). Higher is better except MAE. CoGAR values are means over ten cross-validation seeds with standard deviation. All other methods are deterministic and are reported as single values.²

Feature family	QWK
Entity-relation graph	0.023 ± 0.005
Named entities (NER)	0.338 ± 0.002
AARI (length only)	0.622 ± 0.002
Co-occurrence <i>volume</i> (size)	0.682 ± 0.001
Lexical (morphology, rarity, length)	0.696 ± 0.003
Graph (cooc.+NER+rel.)	0.748 ± 0.004
All features	0.750 ± 0.005
AARI + all features	0.752 ± 0.004
Co-occurrence graph (vol.+topo.)	0.754 ± 0.002
Co-occurrence <i>topology</i>	0.756 ± 0.001

Table 2: Feature-family ablation on BAREC (ordinal LogisticAT, five-fold CV), reported as mean $\pm$ standard deviation over ten seeds. Topology dominates size and entity/relation graphs fail. Topology and the full feature set are statistically indistinguishable (paired bootstrap, $p = 0.52$), so no configuration is marked as best.

tablished. The two CoGAR variants are equivalent ($p = 0.52$), which supports adopting the simpler topology configuration. The classical formulas remain far behind every learned method ($p < 0.0001$).

6.6 Distillation results

Table 3 reports the distilled NeoAraBERT. Trained only on Gemini silver labels, without ever seeing a human label, the student transfers to human BAREC labels at QWK 0.714 with the focal loss, above the class-weighted variant (0.708) and competitive with the arabertv02-word-CE classifier (0.694) and Fanar (0.702). The NeoAraBERT row in Table 1 shows the focal variant. Both variants are reported in Table 3. Its confusion matrix (Figure 1) shows that errors concentrate on the ± 1 band, with a systematic under-prediction of the hardest level (most level-4 paragraphs are predicted as level-3).

Loss	Exact	± 1	MAE	QWK
Weighted CE	0.407	0.912	0.682	0.708
Focal	0.443	0.893	0.666	0.714

Table 3: NeoAraBERT distilled from Gemini silver labels, evaluated on human-labelled BAREC. We report both loss variants rather than selecting one. The gap between them is 0.006 QWK, which we do not treat as meaningful. The comparison was made on BAREC rather than on a held-out set.

6.7 Combining the graph and the LLM

Averaging GPT-4.1 with the co-occurrence topology (*fusion*) gives QWK 0.768, which does *not* beat GPT-4.1 alone (0.784). However, *stacking* an ordinal meta-learner on the GPT-4.1 rating and the graph topology features reaches $\text{QWK} = 0.839 \pm 0.001$ (5 seeds $\times$ 5 folds), well above either component (Table 4); the protocol and its leakage controls are detailed below.

Stacking protocol. Our strongest system combines the LLM and the graph with an ordinal meta-learner: an ordinal logistic regression (LogisticAT) fed two complementary inputs, the GPT-4.1 zero-shot rating and the co-occurrence *topology* features of the document. We guard against leakage on every input. First, GPT-4.1 is used zero-shot and is never trained on our data, so its rating carries no label information. Second, the topology inputs are deterministic functions of the text (graph density, clustering, path length, degree entropy, assortativity), not the output of any trained model. Third, the meta-learner is evaluated strictly out-of-fold: under stratified five-fold cross-validation, each paragraph receives its stacked prediction from a meta-model trained only

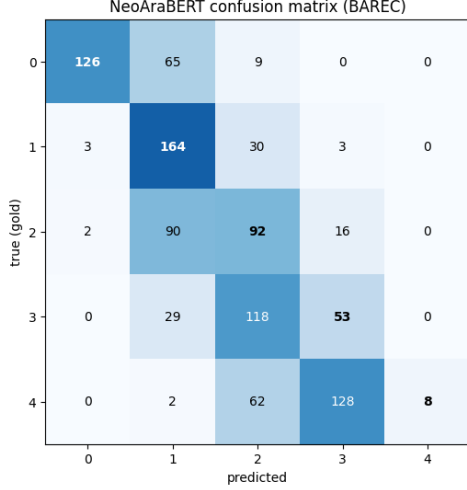


Figure 1: NeoAraBERT (focal) confusion matrix on BAREC (rows: gold, columns: predicted). Errors are concentrated on the ± 1 band; the hardest level is systematically under-predicted.

System	QWK
Co-occurrence topology (alone)	0.756 ± 0.001
GPT-4.1 (alone)	0.784
Qwen-35B (alone)	0.751
Fusion (averaging GPT-4.1 and topology)	0.768
Stacking (GPT-4.1 + topology)	0.839 ± 0.001
Stacking (Qwen-35B + topology)	0.852 ± 0.002

Table 4: Combining the LLM and the graph. Naive averaging does not help but stacking is complementary. The gain does not depend on a proprietary model: stacking the open-weight Qwen-35B with the topology features reaches a slightly higher score than stacking GPT-4.1.

on the other folds, so no paragraph is ever used to both fit and score the meta-learner. As the meta-learner has no tuned hyper-parameters, a single (non-nested) cross-validation level is unbiased. We repeat the procedure over five random seeds and report mean $\pm$ standard deviation, $\text{QWK} = 0.839 \pm 0.001$. Replacing the topology features by the topology *prediction* alone yields a comparable $\text{QWK} = 0.836$, confirming the gain is not an artefact of one particular meta-feature choice. The same procedure with Qwen-35B, an open-weight model, reaches $\text{QWK} = 0.852 \pm 0.002$, slightly above the GPT-4.1 stacking. The graph and the LLM therefore capture *complementary* signals. The gain does not depend on a proprietary model.

Evaluation without level 0. Since level 0 comes from an external source and is trivially separable,

we also report performance on the four BAREC levels alone, with CoGAR retrained on those levels only. Every learned method loses ground: GPT-4.1 falls from 0.784 to 0.604, Qwen-35B from 0.751 to 0.584, Fanar from 0.702 to 0.565 and CoGAR topology from 0.756 to 0.429. This drop is largely mechanical. QWK is a ratio of observed to expected disagreement. Removing a perfectly separable class at one end of the scale sharply reduces the expected disagreement without reducing the observed one. Within-1 accuracy, which is more informative on an ordinal scale, remains high and tightly grouped: 0.901 for GPT-4.1, 0.891 for Qwen-35B, 0.896 for Fanar and 0.863 for CoGAR topology. The surface formulas retain almost no signal in this restricted setting, which indicates that what they capture is concentrated at the easiest end of the scale.

7 Cross-source Distributional Analysis

Figure 2 shows, for each of the seven sources, how eight methods distribute the five reading levels, together with their mean predicted level. This is run on unlabelled data, so it probes *generalisation* rather than accuracy. We compare each method’s mean predicted level per source against the intuitive order children’s stories < detective stories < middle-school < high-school < newspaper < research books, using Kendall’s τ . Arabic Wikipedia is reported separately rather than placed in this order: its articles range from short factual entries to technical texts. All eight methods place it among the easiest sources, which is consistent with the brevity of many Arabic Wikipedia entries. CoGAR and the LLMs follow the intuitive trend (τ from 0.73 to 0.87, CoGAR at 0.83), while the formulas do not (OSMAN 0.47, AARI 0.33, Flesch 0.07). No method recovers the exact order, so the agreement concerns the overall trend rather than the precise ranking. Power is limited with six sources. This confirms, on data with no gold labels, that the structural and LLM-based methods carry a transferable notion of difficulty, whereas surface formulas do not.

8 Conclusion

We showed that CoGAR, a simple and interpretable co-occurrence *network topology* method, predicts Arabic reading difficulty as well as a 35B-parameter (3B active) LLM and well above the surface formulas Flesch and OSMAN, with the signal

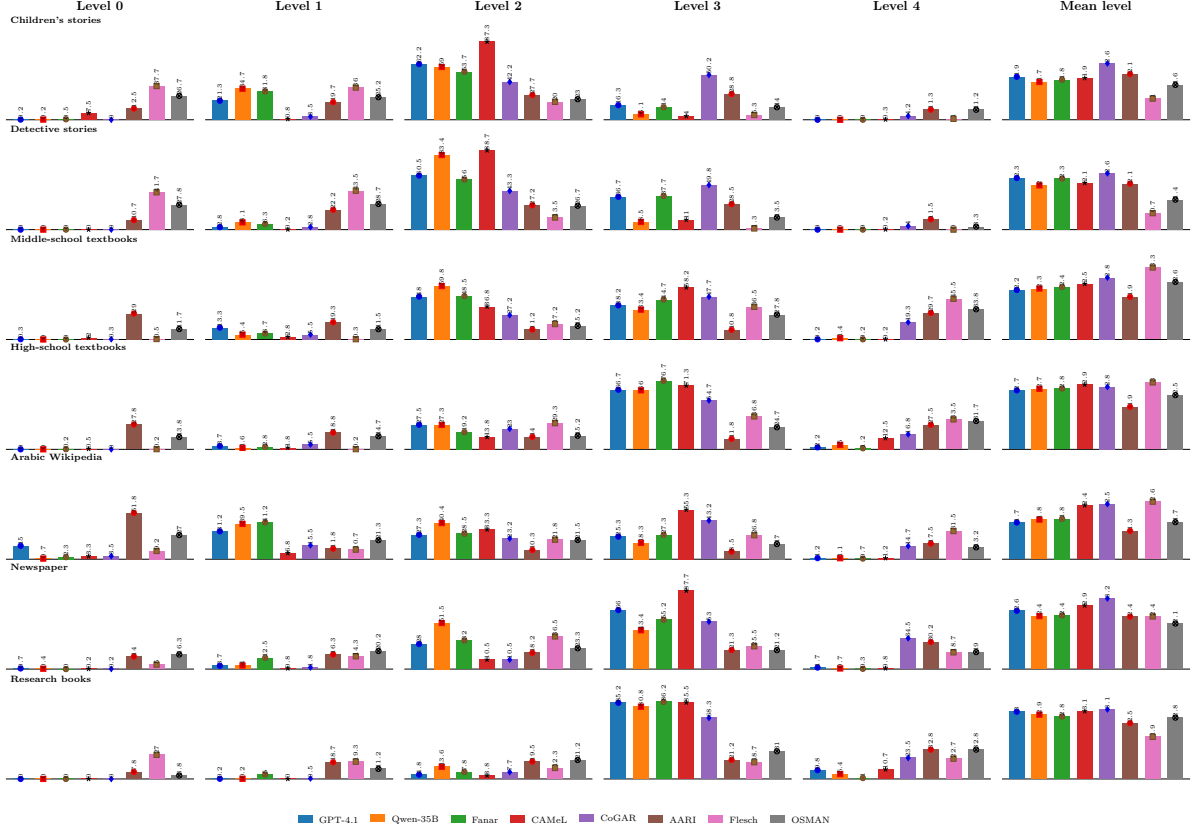


Figure 2: Distribution of predicted reading levels (and mean level, rightmost column) for eight methods across seven Arabic text sources. CoGAR and the LLMs follow the intuitive difficulty trend (Kendall’s τ between 0.73 and 0.87 over six sources). Flesch and OSMAN are noisy or inverted.

residing in the graph’s shape rather than its size. We distilled a frontier LLM into a small Arabic encoder. We found that stacking an LLM with CoGAR yields our best result, whether the LLM is proprietary or open-weight, the two signals being complementary. A cross-source analysis further showed that structural and LLM-based methods generalise to unlabelled text where formulas fail. Future work includes extending to more corpora, exploring lighter open models as graph companions and evaluating whether topology still dominates volume at BAREC’s native sentence granularity, where very short texts yield nearly complete graphs.

9 Limitations

Supervised evaluation relies on a single corpus (BAREC); broader corpora would strengthen the conclusions. The easiest level (level 0) is drawn entirely from a separate first-grade textbook rather than from BAREC; its much shorter sentences make it more easily separable, so absolute scores may be partly inflated by this class, and excluding

this class changes the relative ranking: CoGAR then scores below the three evaluated zero-shot LLMs. Enriching this level with additional easy-text sources is left to future work. The LLM raters are closed and costly. We cannot rule out exposure to similar data during pre-training, which is precisely an argument in favor of an LLM-free method whose signal provenance is verifiable: the CoGAR features are deterministic functions of the text, computable and inspectable. The co-occurrence graph depends on the quality of SinaTools’ morphological analysis. Its window size proved robust (Table 6) but its feature set is only lightly tuned. The distillation teacher labels are themselves model-generated (silver), so the student inherits any teacher bias. Finally, the cross-source distributional study has no gold labels and is therefore qualitative.

References

Chadi Abou Chakra, Hadi Khaled Hamoud, Osama Rakan Al Mraikhat, Qusai Abu Obaida, Mohamad Ballout, and Fadi Zaraket. 2026. [NeoAraBERT: A](#)

- modern foundation model for Arabic embeddings with diacritics-aware tokenization and POS-targeted masking. In *Findings of the Association for Computational Linguistics: ACL 2026*, pages 25952–25968, San Diego, California, United States. Association for Computational Linguistics.
- Shahad Al-Khalifa, Nadir Durrani, Hend Al-Khalifa, and Firoj Alam. 2025. [The landscape of arabic large language models \(allms\): A new era for arabic language technology](#). *Preprint*, arXiv:2506.01340.
- Muhammed Al Khalil, Nizar Habash, and Zhengyang Jiang. 2020. [A large-scale leveled readability lexicon for Standard Arabic](#). In *Proceedings of the Twelfth Language Resources and Evaluation Conference*, pages 3053–3062, Marseille, France. European Language Resources Association.
- Abdel Karim Al-Tamimi, Manar Jaradat, Nuha Aljarrah, and Sahar Ghanim. 2014. AARI: Automatic Arabic readability index. *International Arab Journal of Information Technology*, 11(4):370–378.
- Bashar Alhafni, Reem Hazim, Juan David Pineros Liberato, Muhammed Al Khalil, and Nizar Habash. 2024. [The SAMER Arabic text simplification corpus](#). In *Proceedings of the 2024 Joint International Conference on Computational Linguistics, Language Resources and Evaluation (LREC-COLING 2024)*, pages 16079–16093, Torino, Italia. ELRA and ICCL.
- Bertram Bruce, Andee Rubin, and Kathleen Starr. 1981. Why readability formulas fail. *IEEE Transactions on Professional Communication*, (1):50–52.
- Jacob Cohen. 1968. Weighted kappa: Nominal scale agreement provision for scaled disagreement or partial credit. *Psychological bulletin*, 70(4):213.
- BAK Dawood. 1977. The relationship between readability and selected language variables. Unpublished master’s thesis (in arabic), Baghdad University, Iraq.
- Mahmoud El-Haj and Paul Rayson. 2016. [OSMAN — a novel Arabic readability metric](#). In *Proceedings of the Tenth International Conference on Language Resources and Evaluation (LREC’16)*, pages 250–255, Portorož, Slovenia. European Language Resources Association (ELRA).
- Mo El-Haj, Sultan Almujaivel, Damith Premasiri, Tharindu Ranasinghe, and Ruslan Mitkov. 2024. [DARES: Dataset for Arabic readability estimation of school materials](#). In *Proceedings of the Workshop on DeTermIt! Evaluating Text Difficulty in a Multilingual Context @ LREC-COLING 2024*, pages 103–113, Torino, Italia. ELRA and ICCL.
- Khalid N. Elmadani, Bashar Alhafni, Hanada Taha, and Nizar Habash. 2025a. [BAREC shared task 2025 on Arabic readability assessment](#). In *Proceedings of The Third Arabic Natural Language Processing Conference: Shared Tasks*, pages 239–252, Suzhou, China. Association for Computational Linguistics.
- Khalid N. Elmadani, Nizar Habash, and Hanada Taha-Thomure. 2025b. [A large and balanced corpus for fine-grained Arabic readability assessment](#). In *Findings of the Association for Computational Linguistics: ACL 2025*, pages 16376–16400, Vienna, Austria. Association for Computational Linguistics.
- Khalid N. Elmadani, Adel Mahmoud Wizani, Hanada Taha Thomure, and Nizar Habash. 2026. [A large and balanced multi-domain arabic corpus annotated for morphology, syntax, and readability](#). In *Proceedings of the Fifteenth Language Resources and Evaluation Conference (LREC 2026)*, pages 11761–11775, Palma, Mallorca, Spain. European Language Resources Association (ELRA).
- Lijun Feng, Martin Jansche, Matt Huenerfauth, and Noémie Elhadad. 2010. [A comparison of features for automatic readability assessment](#). In *Coling 2010: Posters*, pages 276–284, Beijing, China. Coling 2010 Organizing Committee.
- Rudolf Flesch. 1948. [A new readability yardstick](#). *Journal of Applied Psychology*, 32(3):221–233.
- Gemini Team, Google DeepMind. 2026. Gemini 3.5 Flash. Generally available May 2026. <https://deepmind.google/models/gemini/flash/>.
- Robert Gunning. 1969. The fog index after twenty years. *Journal of Business Communication*, 6(2):3–13.
- Nizar Habash, Hanada Taha-Thomure, Khalid N. Elmadani, Zeina Zeino, and Abdallah Abushmaes. 2025. Guidelines for fine-grained sentence-level arabic readability annotation. In *Proceedings of the 19th Linguistic Annotation Workshop (LAW-XIX)*, Vienna, Austria. Association for Computational Linguistics.
- Tymaa Hammouda, Mustafa Jarrar, and Mohammed Khalilia. 2024. SinaTools: Open source toolkit for Arabic natural language understanding. In *Proceedings of the 2024 AI in Computational Linguistics (ACLING 2024)*, Procedia Computer Science, Dubai, United Arab Emirates. Elsevier.
- Mustafa Jarrar, Mohammed Khalilia, and Sana Ghanem. 2022. [Wojood: Nested Arabic named entity corpus and recognition using BERT](#). In *Proceedings of the Thirteenth Language Resources and Evaluation Conference*, pages 3626–3636, Marseille, France. European Language Resources Association.
- Nouran Khallaf and Serge Sharoff. 2021. [Automatic difficulty classification of Arabic sentences](#). In *Proceedings of the Sixth Arabic Natural Language Processing Workshop*, pages 105–114, Kyiv, Ukraine (Virtual). Association for Computational Linguistics.
- J. Peter Kincaid, Robert P. Fishburne, Richard L. Rogers, and Brad S. Chissom. 1975. Derivation of

- new readability formulas (Automated Readability Index, Fog Count, and Flesch Reading Ease Formula) for Navy enlisted personnel. Technical report, Naval Technical Training Command, Millington, TN, Research Branch.
- Juan Liberato, Bashar Alhafni, Muhamed Khalil, and Nizar Habash. 2024. [Strategies for Arabic readability modeling](#). In *Proceedings of the Second Arabic Natural Language Processing Conference*, pages 55–66, Bangkok, Thailand. Association for Computational Linguistics.
- G. Harry McLaughlin. 1969. SMOG grading: A new readability formula. *Journal of Reading*, 12(8):639–646.
- Ministry of Education, Kingdom of Saudi Arabia. 2024. *Lughati lil-Saff al-Awwal al-Ibtida’i: al-Fasl al-Dirasi al-Awwal*, 1 edition. Ministry of Education, Riyadh, Saudi Arabia.
- Naoual Nassiri, Violetta Cavalli-Sforza, and Abdelhak Lakhouaja. 2020. [Mosar: Modern standard arabic readability corpus for ll learners](#). In *Proceedings of the 4th International Conference on Big Data and Internet of Things, BDIoT ’19*, New York, NY, USA. Association for Computing Machinery.
- Naoual Nassiri, Violetta Cavalli-Sforza, and Abdelhak Lakhouaja. 2023. [Approaches, methods, and resources for assessing the readability of arabic texts](#). *ACM Transactions on Asian and Low-Resource Language Information Processing*, 22(4).
- Naoual Nassiri, Abdelhak Lakhouaja, and Violetta Cavalli-Sforza. 2022. [Combining classical and non-classical features to improve readability measures for arabic first language texts](#). In *Advanced Intelligent Systems for Sustainable Development (AI2SD’2020)*, pages 463–470.
- OpenAI. 2025. Introducing GPT-4.1 in the API. <https://openai.com/index/gpt-4-1/>. Accessed: 2025.
- Fabian Pedregosa, Francis Bach, and Alexandre Gramfort. 2017. [On the consistency of ordinal regression methods](#). *Journal of Machine Learning Research*, 18(55):1–35.
- Xinying Qiu, Yuan Chen, Hanwu Chen, Jian-Yun Nie, Yuming Shen, and Dawei Lu. 2021. [Learning syntactic dense embedding with correlation graph for automatic readability assessment](#). In *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing (Volume 1: Long Papers)*, pages 3013–3025, Online. Association for Computational Linguistics.
- Qwen Team. 2026. [Qwen3.5: Towards native multi-modal agents](#). Alibaba Cloud technical blog. Accessed 2026.
- Janice Redish. 2000. Readability formulas have even more limitations than Klare discusses. *ACM Journal of Computer Documentation*, 24(3):132–137.
- Hind Saddiki, Karim Bouzoubaa, and Violetta Cavalli-Sforza. 2015. [Text readability for Arabic as a foreign language](#). In *2015 IEEE/ACS 12th International Conference of Computer Systems and Applications (AICCSA)*, pages 1–8.
- Hind Saddiki, Nizar Habash, Violetta Cavalli-Sforza, and Muhamed Al Khalil. 2018. [Feature optimization for predicting readability of Arabic L1 and L2](#). In *Proceedings of the 5th Workshop on Natural Language Processing Techniques for Educational Applications*, pages 20–29, Melbourne, Australia. Association for Computational Linguistics.
- FANAR TEAM, Ummar Abbas, Mohammad Shahmeer Ahmad, Minhaj Ahmad, Abdulaziz Al-Homaid, Anas Al-Nuaimi, Enes Altinisik, Ehsaneddin Asgari, Sanjay Chawla, Shammur Chowdhury, Fahim Dalvi, Kareem Darwish, Nadir Durrani, Mohamed Elfeky, Ahmed Elmagarmid, Mohamed Eltabakh, Asim Ersoy, Masoomali Fatehkia, Mohammed Qusay Hashim, and 18 others. 2026. [Fonar 2.0: Arabic generative ai stack](#).
- Yang Zhang, Amr Mohamed, Hadi Abdine, Guokan Shang, and Michalis Vazirgiannis. 2026. Beyond random sampling: Efficient language model pretraining via curriculum learning. In *Proceedings of the 19th Conference of the European Chapter of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 5776–5794.

A Full Feature Inventory

Table 5 lists the 40 features in the four families used in Section 4, with the volume and topology split that Table 2 scores. Volume features count the size of the graph. Topology features describe its shape independently of that size.

B Co-occurrence Window Sensitivity

Table 6 reports the topology configuration for co-occurrence window sizes from one to five content words, over five seeds per value.

C Data Provenance, Versions and Licenses

Table 7 documents the origin of every dataset used in this work. The unlabelled corpora were drawn from the collection of the Arab Center for Research and Policy Studies³. None of the collections contains personal data.

³<https://www.dohainstitute.org>

Models and versions. SinaTools 1.0.9 (Ham-mouda et al., 2024) for morphological analysis and named entity recognition. The CAMEL readability classifiers readability-arabertv02-word-CE and readability-arabertv2-d3tok-reg (El-madani et al., 2025b), obtained from their public releases. NeoAraBERT (Abou Chakra et al., 2026) as the distillation student. The LLM raters GPT-4.1 (OpenAI, 2025), Qwen-35B (Qwen Team, 2026) and Fanar (TEAM et al., 2026) were queried through their respective APIs. Gemini-3.5-flash (Gemini Team, Google DeepMind, 2026) produced the silver labels.

D LLM Prompt

The same instruction was used for the three zero-shot raters and for the Gemini silver labelling. Everything before `Text to assess:` is sent as the system message and the document follows as the user message. Decoding is greedy with temperature 0. The prompt uses a 1 to 5 scale, which the evaluation harness maps to our 0 to 4 levels.

You are an expert in Arabic linguistics and reading comprehension. Your task is to assess the readability level of a given Arabic text. Use the following 5-level scale: 1 = Very Easy (texts for early primary school, very simple vocabulary and short sentences) 2 = Easy (texts for upper primary school, common everyday vocabulary) 3 = Medium (texts for middle school, moderately complex vocabulary and sentence structure) 4 = Difficult (texts for high school or undergraduate level, advanced vocabulary and complex syntax) 5 = Very Difficult (specialized or academic texts, technical vocabulary, complex argumentation) Rules: Base your judgment on vocabulary complexity, sentence length and structure, and level of abstraction. Do NOT consider the topic or domain alone, focus on linguistic difficulty. Respond with ONLY a single integer: 1, 2, 3, 4 or 5. Nothing else.

Text to assess:

""" [document] """

Family	Feature	Description
<i>Lexical (8)</i>		
	lex_n_tokens	token count
	lex_words_per_sent	tokens per sentence
	lex_avg_word_len	mean token length in characters
	lex_ttr	distinct lemmas over tokens
	lex_content_ttr	distinct content lemmas over content tokens
	lex_root_diversity	distinct roots over tokens
	lex_mean_log_freq	mean log frequency of content lemmas
	lex_rare_ratio	share of content lemmas with corpus frequency below 50,000
<i>Co-occurrence, volume (2)</i>		
	cooc_n_nodes	number of content lemmas in the graph
	cooc_n_edges	number of co-occurrence links
<i>Co-occurrence, topology (10)</i>		
	cooc_avg_degree	mean number of links per lemma
	cooc_max_degree	degree of the most connected lemma
	cooc_density	realised links over possible links
	cooc_avg_clustering	mean clustering coefficient
	cooc_n_components	number of connected components
	cooc_largest_cc_frac	share of nodes in the largest component
	cooc_avg_path	mean shortest path length
	cooc_diameter	longest shortest path
	cooc_degree_entropy	entropy of the degree distribution
	cooc_assortativity	degree correlation between linked lemmas
<i>Named entities (5)</i>		
	ner_n_entities	number of distinct entities
	ner_n_mentions	number of entity mentions
	ner_distinct_cats	number of distinct Wojoood categories
	ner_entities_per_sent	entities per sentence
	ner_cat_entropy	entropy over entity categories
<i>Entity-relation graph (15)</i>		
	rel_n_relations	number of extracted relations
	rel_n_rel_types	number of distinct relation types
	rel_rel_entropy	entropy over relation types
	rel_n_nodes	number of entity nodes
	rel_n_edges	number of relation edges
	rel_avg_degree	mean degree
	rel_max_degree	maximum degree
	rel_density	realised edges over possible edges
	rel_avg_clustering	mean clustering coefficient
	rel_n_components	number of connected components
	rel_largest_cc_frac	share of nodes in the largest component
	rel_degree_entropy	entropy of the degree distribution
	rel_avg_path	mean shortest path length
	rel_diameter	longest shortest path
	rel_assortativity	degree correlation between linked entities

Table 5: The 40 features, by family. The co-occurrence family is split into the two volume features and the ten topology features that Table 2 evaluates separately. The same topology measures are computed on the entity-relation graph, so the two graphs are compared under identical descriptors.

Co-occurrence window	QWK
1	0.764 ± 0.004
2 (ours)	0.756 ± 0.001
3	0.748 ± 0.001
4	0.740 ± 0.002
5	0.745 ± 0.003

Table 6: Sensitivity of the topology configuration to the co-occurrence window, over five seeds per value. The total range is 0.024, which indicates that the method is robust to this parameter rather than dependent on fine tuning.

Dataset	Source
BAREC levels 1–4	Sentence-level release of the BAREC corpus (Elmadani et al., 2025b), used under its public release terms
Level 0	Passages extracted from a first-grade Arabic textbook of the Saudi national curriculum (Ministry of Education, Kingdom of Saudi Arabia, 2024)
Multi-source corpus (600 chunks per source)	Children’s stories, detective stories, middle-school textbooks, high-school textbooks, Arabic Wikipedia, As-Safir newspaper articles and research manuscripts from the Arab Center for Research and Policy Studies.
Distillation corpus (6,000 chunks per source)	Hindawi Foundation open-access literature, school textbooks, Arabic Wikipedia, As-Safir newspaper articles and research manuscripts from the Arab Center for Research and Policy Studies.

Table 7: Provenance of the datasets used in this work.